



Rating
Hold

North America
United States

TMT
Semiconductors

Company
NVIDIA Corporation

Reuters
NVDA.OQ

Bloomberg
NVDA US

Exchange
NMS

Ticker
NVDA

Date
28 August 2024

Forecast Change

Price at 28 Aug 2024 (USD)	125.61
Price target	115.00
52-week range	696.41 - 40.32

F2Q25 (Jul) Results: Strong execution, but high bar

NVDA delivered another substantial >\$1B beat/raise in its F2Q/F3Q top-line report/guide (+6%/+6% respectively vs. DBE), and perhaps more importantly, allayed concerns of a more protracted delay in its next-gen Blackwell family of chips by guiding for shipments of "several billion dollars" in F4Q. The strong top-line result/guidance were powered by accelerated purchases of Hopper, with demand strength importantly expected to continue even as Blackwell hits the market (growth accelerating in F4Q?). The primary negative of the print/guide/Blackwell commentary comes on the GM line where NVDA reiterated that rising mix of new products would continue to weigh on GM into F3Q and seemingly even more so in F4Q (as Blackwell truly ramps). Overall, we remain impressed with NVDA's technological leadership in the exploding AI sector, and despite the smallest of hiccups in its gen-to-gen transition (seemingly weighing on share price in after market), believe Blackwell will ramp strongly and the co's market dominance will continue into CY25+. However, even with this positive fundamental view yielding a +~15% increase in our CY25 EPS ests, based on a consistent 30x P/E multiple (and resulting \$115 P/T), we view the shares as fairly valued. Maintain Hold.

Positives outweigh negatives

Positives: **1)** NVDA continues to benefit from strong and diversified demand (evidenced by more than 50% of Data Center revs coming from consumer internet and enterprise companies), delivering revenues of ~\$30b (vs. DBE of \$28.5b) with Data Center once again leading the way on continued AI goodness (specifically focused on Hopper purchases). **2)** Despite concerns around a delayed launch of the co's Blackwell platform (due to technical concerns, see note [here](#)), the company expressed confidence that the delays are limited with its expectations to ship "ship several billion dollars" of Blackwell revenue in F4Q (while still seeing strong Hopper demand as well). **3)** After the first half of FY25 the co has returned ~\$15.4b to shareholders (inclusive of repurchases and dividends), and has an additional \$57.5b share repurchase authorization remaining following its board adding \$50b.

Negatives: **1)** The co saw GM declined a significant (albeit expected) -320bps q/q in F2Q due to inventory provisions for low-yielding Blackwell material and a mix shift towards new products within the co's Data Center business. This trend is expected to directionally continue in F3Q (GM guided down -70bps q/q) and again in F4Q (implied in ~-200bps q/q) as new product mix rises/accelerates. **2)** Opex continues to grow at a significant clip, up +12% q/q and 52% y/y, albeit accompanying continued aggressive revenue expansion (with F2Q revs up +15%

Valuation & Risks

Ross Seymore

Research Analyst
+1-415-617-3268

Melissa Weathers

Research Analyst
+1-212-250-2134

DJ Sebastian

Research Associate
+1-415-262-2007

Gianmarco Vella

Research Associate
+1-857-891-9383

Key changes

TP	100.00 to 115.00	↑	15.0%
EPS (USD)	2.46 to 2.72	↑	10.8%
Revenue (USDm)	118,176.0 to 126,461.5	↑	7.0%

Source: Deutsche Bank

Deutsche Bank Securities Inc.

IMPORTANT RESEARCH DISCLOSURES AND ANALYST CERTIFICATIONS LOCATED IN APPENDIX 1. Deutsche Bank does and seeks to do business with companies covered in its research reports. Thus, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. MCI (P) 041/10/2023.

28 August 2024
Semiconductors
NVIDIA Corporation



q/q and +122% y/y). FY25 opex growth guidance increased to +mid-to-high 40% range y/y (to ~\$11.5b vs ~\$11.3b prior), but even with this increase the company should yield better opex intensity than we formerly modeled.

Another significant beat and raise

NVDA reported Jul-qtr revenues of \$30.0b (+15% q/q, +122% y/y), once again solidly above guidance/DBe of \$28.0b/\$28.4b and Street at \$28.8b. By end market, Data Center revs beat our expectations (+\$1.6b above DBe) led by growth in Enterprise. Gaming revs were below our expectations (+8% q/q vs DBe +10% q/q), Proviz revs of \$454m (+6% q/q vs. DBe +5% q/q) were above our expectations, Auto revs of \$346m (+5% q/q vs. DBe +15% q/q) were below our ests, and OEM/Other revs of \$88m beat our ests at \$82m. PF GMs came in at 75.7%, vs. guidance 75.5%, and PF Opex came in at \$2.79b vs. guidance/DBe \$2.80b. All told, Jul-qtr EPS totaled \$0.68, above DBe/Street at \$0.63/\$0.65 on better revs & margins. NVDA guided revenues to \$32.5b (+8.2% q/q), above DBe/Street at \$32.3b/\$31.9b on continued strength in Data Center. Non-GAAP GM was guided to 75.0% (vs. DBE 74.6%), with non-GAAP OpEx guided to \$3.0b (vs. DBE \$2.94b), yielding implied non-GAAP EPS of ~\$0.74, above DBE/Street at \$0.68/\$0.71. Our CY24E revenue/ EPS moves to \$126.5b (+108% y/y)/ \$2.85 from \$118.1b (+94% y/y)/ \$2.65, and our CY25E revs/ EPS move to \$173.1b (+37% y/y)/ \$3.85 from \$146.2b (+24% y/y)/ \$3.29 prior. Our CY26E rev/ EPS ests move to \$195.1b (+13% y/y)/ \$4.40 from \$161.9b (+11% y/y)/ \$3.68 prior.

Maintain Hold rating, raise P/T to \$115

On higher estimates, we raise our P/T to \$115 and maintain our Hold rating. Our P/T is based on a consistent multiple of ~30x our CY25E EPS, a premium to the co's recent trading range in the 20-25x range to reflect strong earnings momentum, but still a discount to the co's 5-year average. Upside/downside risks include better-/ worse-than-expected Gaming, Data Center, Automotive, and OEM/Other revenue growth, higher-/ lower-than-expected profitability, macro, design win/product execution, and market share shifts. Positive/negative company-specific risks include market share gains/losses in Graphics and Compute & Networking businesses, competitive and pricing pressure, product execution, and impacts to profitability from investments.

28 August 2024
Semiconductors
NVIDIA Corporation



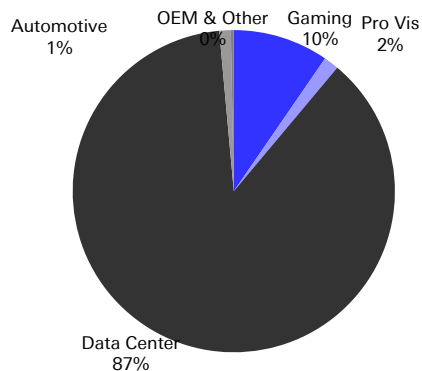
Details

Figure 1: NVIDIA earnings result summary

	Consensus	2025 DBE	2025 Actual	Comments
Revenue	\$28,857	\$28,472	\$30,040	Above guidance/DBE of \$28.0b/\$28.4b
q/q	+11%	+9%	+15%	
Gross Margin		75.5%	75.7%	
Opex		\$2,805	\$2,792	Above DB est
Tax Rate		17.0%	16.6%	
EPS	\$0.64	\$0.63	\$0.68	Above DB est
End Market Segment Results				
Gaming		\$2,927	\$2,880	
q/q		+11%	+9%	Below DB est
Professional Visualization		\$448	\$454	
q/q		+5%	+6%	Above DB est
Datacenter		\$24,636	\$26,272	
q/q		+9%	+16%	Above DB est
Automotive		\$378	\$346	
q/q		+15%	+5%	Below DB est
OEM & Other		\$82	\$88	
q/q		+5%	+13%	
Balance Sheet				
Cash and equivalents		\$39,786	\$34,800	
q/q		\$8,348	\$3,362	
Inventory		\$7,943	\$6,675	
q/q		+\$2079	+\$811	
DIOs		104 days	83 days	
q/q		+6 days	-14 days	
Cash Flow Statement				
CFO		\$14,063	\$14,489	
Capex		\$469	\$977	
FCF		\$13,594	\$13,512	

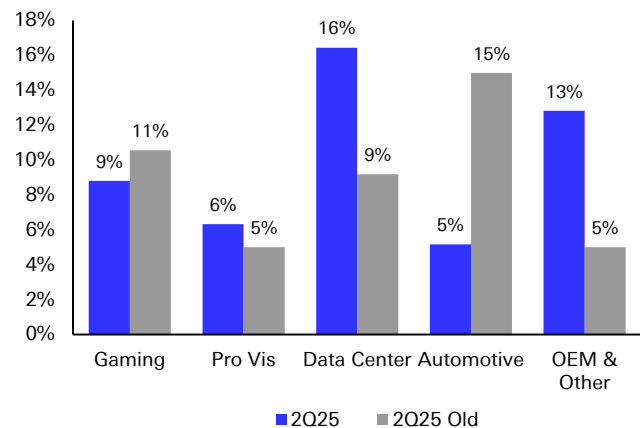
Source : Company data, Bloomberg Finance LP, Deutsche Bank estimates

Figure 2: Reported quarter product mix



Source : Company data, Deutsche Bank estimates

Figure 3: Reported quarter q/q change (Actual vs. DBE)



Source : Company data, Deutsche Bank estimates

28 August 2024
Semiconductors
NVIDIA Corporation



Guidance and estimate revisions

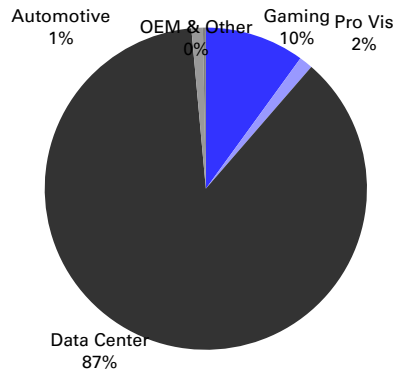
NVIDIA provided the following guidance:

Figure 4: NVIDIA quarterly guidance

Guidance item	Guidance	Prior DBe
Revenues	\$32.5b +/-2%	--
q/q change (implied)	+6-10% q/q	+7%
Revenue Midpoint (\$ in b)	\$32.50	\$30.60
Non-GAAP GM%	75% +/- 50bps	74.6%
PF opex (m)	\$3,000	\$2,939
Other income/(expense)	\$350m income	\$396m income
Tax rate	17.0% +/- 1%	17.0%
Gaming	--	11%
Datacenter	--	7%
Automotive	--	20%
Provisualization	--	0%

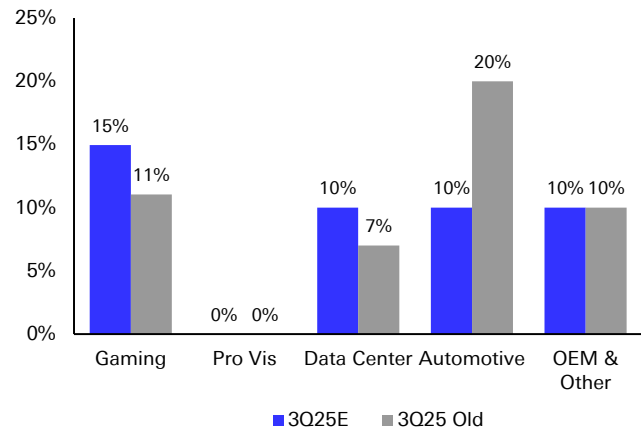
Source : Company reports, Deutsche Bank estimates

Figure 5: Fwd qtr est product segment mix



Source : Deutsche Bank estimates

Figure 6: Fwd qtr est segment growth (new vs. old)



Source : Deutsche Bank estimates

Given NVDA's guidance, we adjust our estimates as illustrated below:

28 August 2024
Semiconductors
NVIDIA Corporation



Figure 7: NVIDIA estimates summary

	<u>New 3Q25E</u>	<u>Old 3Q25E</u>	<u>New CY24E</u>	<u>Old CY24E</u>	<u>New CY25E</u>	<u>Old CY25E</u>
Revenue	\$33,141	\$30,603	\$126,461	\$118,176	\$173,072	\$146,225
q/q	+10%	+7%	--	--	--	--
y/y	+83%	+69%	+108%	+94%	+37%	+24%
Gross Margin	75.0%	74.6%	75.4%	75.6%	73.0%	74.1%
EPS	\$0.74	\$0.68	\$2.85	\$2.65	\$3.85	\$3.29

Source : Deutsche Bank estimates

Figure 8: NVIDIA Income Statement (\$ in millions, except per share data)

	Apr-23 1Q24	Jul-23 2Q24	Oct-23 3Q24	Jan-24 4Q24	Year 2024	Apr-24 1Q25	Jul-24 2Q25	Oct-24 3Q25	Jan-25 4Q25	Year 2025E	Apr-25 1Q26E	Jul-25 2Q26E	Oct-25 3Q26E	Jan-26 4Q26E	Year 2026E	Apr-26 1Q27E	Jul-26 2Q27E	Oct-26 3Q27E	Jan-27 4Q27E	Year 2027E
Revenues	\$7,192	\$13,507	\$18,120	\$22,103	\$60,922	\$26,044	\$30,040	\$33,141	\$37,236	\$126,461	\$40,134	\$42,234	\$44,630	\$46,073	\$173,072	\$46,025	\$47,743	\$50,218	\$51,079	\$195,065
Depreciation	211	228	237	250	926	270	289	294	299	1,152	304	309	314	319	1,248	324	329	334	339	1,326
Cost of Goods	2,179	3,665	4,300	4,894	15,038	5,214	7,022	7,987	9,768	29,991	10,686	11,287	11,567	11,885	45,425	12,071	12,677	13,600	14,001	52,350
Gross Profit (Non-GAAP)	4,802	9,614	13,583	16,959	44,958	20,560	22,729	24,860	27,169	95,319	29,144	30,637	32,750	33,889	126,401	33,629	34,736	36,284	36,739	141,389
Research and Development (Non-GAAP)	1,339	1,440	1,581	1,747	6,107	1,981	2,241	2,390	2,582	9,194	2,743	2,887	3,051	3,149	11,830	3,275	3,287	3,407	3,466	13,375
Selling, General and Admin. (Non-GAAP)	411	398	445	463	1,717	520	551	608	646	2,325	676	711	752	776	2,915	775	804	846	860	3,285
Other Non-GAAP Adjustments	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Other Operating Exp. / (Inc.)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total Operating Expenses (Non-GAAP)	1,750	1,838	2,026	2,210	7,824	2,501	2,792	2,997	3,228	11,518	3,419	3,598	3,802	3,925	14,744	3,990	4,091	4,253	4,326	16,660
Operating Income	3,052	7,776	11,557	14,749	37,134	18,059	19,937	21,863	23,941	83,800	25,725	27,039	28,948	29,944	111,656	29,639	30,645	32,031	32,413	124,729
Interest Income	150	187	234	294	865	359	444	413	540	1,756	683	858	1,045	1,269	3,855	1,429	1,652	1,870	2,115	7,065
Interest Expense on ST Debt	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Interest Expense on LT Debt	66	65	63	63	257	64	61	61	61	247	61	61	61	61	244	61	61	61	61	244
Interest Expense on Conv. Debt	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total Interest Expense	66	65	63	63	257	64	61	61	61	247	61	61	61	61	244	61	61	61	61	244
Other Non-oper. Exp. / (Inc.)	0	2	(4)	(1)	(3)	(7)	3	0	0	(4)	0	0	0	0	0	0	0	0	0	0
Pretax Income	3,136	7,896	11,732	14,981	37,745	18,361	20,317	22,215	24,420	85,313	26,347	27,837	29,932	31,153	115,268	31,007	32,235	33,840	34,467	131,550
Income Tax Expense / (Benefit)	423	1,156	1,712	2,142	5,433	3,123	3,365	3,777	4,151	14,416	4,479	4,732	5,088	5,296	19,596	5,271	5,480	5,753	5,859	22,363
Non-GAAP Net Income for Diluted EPS	\$2,713	\$6,740	\$10,020	\$12,839	\$32,312	\$15,238	\$16,952	\$18,439	\$20,269	\$70,897	\$21,868	\$23,104	\$24,843	\$25,857	\$95,672	\$25,736	\$26,755	\$28,087	\$28,608	\$109,186
PF Basic EPS	\$0.11	\$0.27	\$0.41	\$0.52	\$1.31	\$0.62	\$0.69	\$0.75	\$0.82	\$2.88	\$0.89	\$0.94	\$1.01	\$1.05	\$3.89	\$1.05	\$1.09	\$1.14	\$1.16	\$4.44
PF Fully Diluted EPS	\$0.11	\$0.27	\$0.40	\$0.52	\$1.30	\$0.61	\$0.68	\$0.74	\$0.82	\$2.85	\$0.88	\$0.93	\$1.00	\$1.04	\$3.85	\$1.04	\$1.08	\$1.13	\$1.15	\$4.40
Weighted Average Shares Out. (in '000s)	24,700,000	24,730,000	24,680,000	24,660,000	24,692,500	24,620,000	24,578,000	24,576,000	24,574,000	24,587,000	24,572,500	24,571,000	24,569,500	24,568,000	24,570,250	24,566,500	24,565,000	24,563,500	24,562,000	24,564,250
Diluted Shares (in '000s)	24,900,000	24,990,000	24,940,000	24,900,000	24,932,500	24,890,000	24,848,000	24,846,000	24,844,000	24,857,000	24,842,500	24,841,000	24,839,500	24,838,000	24,840,250	24,836,500	24,835,000	24,833,500	24,832,000	24,834,250
Net ESO Expense	\$735	\$842	\$979	\$993	\$3,549	\$1,011	\$1,154	\$1,269	\$1,269	\$4,704	\$1,333	\$1,400	\$1,469	\$1,469	\$5,671	\$1,543	\$1,620	\$1,701	\$1,701	\$6,565
Adjusted Basic GAAP EPS (incl-ESO)	\$0.08	\$0.24	\$0.37	\$0.48	\$1.16	\$0.58	\$0.64	\$0.70	\$0.77	\$2.69	\$0.84	\$0.88	\$0.95	\$0.99	\$3.66	\$0.98	\$1.02	\$1.07	\$1.10	\$4.18
Adjusted Diluted GAAP EPS (incl-ESO)	\$0.08	\$0.24	\$0.36	\$0.48	\$1.15	\$0.57	\$0.64	\$0.69	\$0.76	\$2.66	\$0.83	\$0.87	\$0.94	\$0.98	\$3.62	\$0.97	\$1.01	\$1.06	\$1.08	\$4.13
Common Dividends Per Share	0.00	0.00	0.00	0.00	\$0.02	\$0.00	\$0.01	\$0.01	\$0.01	\$0.03	\$0.01	\$0.01	\$0.01	\$0.01	\$0.04	\$0.01	\$0.01	\$0.01	\$0.01	\$0.04
Nonrecurring Charges	\$192	\$73	\$231	(\$118)	\$378	\$71	(\$51)	(\$51)	(\$51)	(\$82)	(\$51)	(\$51)	(\$51)	(\$51)	(\$204)	(\$51)	(\$51)	(\$51)	(\$51)	(\$204)
Tax Effect of Charges	(257)	(363)	(433)	(321)	(1,374)	(725)	(750)	9	9	(1,458)	9	9	9	9	35	9	9	9	9	35
GAAP Net Inc. for Dil. EPS	\$2,043	\$6,188	\$9,243	\$12,285	\$29,759	\$14,881	\$16,599	\$17,212	\$19,042	\$67,733	\$20,577	\$21,747	\$23,416	\$24,429	\$90,170	\$24,236	\$25,178	\$26,428	\$26,949	\$102,790
GAAP Basic EPS	\$0.08	\$0.25	\$0.37	\$0.50	\$1.21	\$0.60	\$0.68	\$0.70	\$0.77	\$2.75	\$0.84	\$0.89	\$0.95	\$0.99	\$3.67	\$0.99	\$1.02	\$1.08	\$1.10	\$4.18
GAAP Diluted EPS	\$0.08	\$0.25	\$0.37	\$0.49	\$1.19	\$0.60	\$0.67	\$0.69	\$0.77	\$2.72	\$0.83	\$0.88	\$0.94	\$0.98	\$3.63	\$0.98	\$1.01	\$1.06	\$1.09	\$4.14
EBITDA	3,263	8,002	11,798	15,000	38,063	18,336	20,223	22,157	24,240	84,956	26,029	27,348	29,262	30,263	112,902	29,963	30,974	32,365	32,752	126,055
EBIT	3,052	7,774	11,561	14,750	37,137	18,066	19,934	21,863	23,941	83,804	25,725	27,039	28,948	29,944	111,656	29,639	30,645	32,031	32,413	124,729
TTM Revenues	25,878	32,681	44,870	60,922	60,922	79,774	96,307	111,328	126,461	126,461	140,552	152,745	164,234	173,072	173,072	178,962	184,471	190,059	195,065	195,065
TTM EBITDA	9,032	15,503	25,532	38,063	38,063	53,136	65,357	75,716	84,956	84,956	92,649	99,775	106,879	112,902	112,902	116,837	120,462	123,566	126,055	126,055
TTM EBIT	8,134	14,580	24,604	37,137	37,137	52,151	64,311	74,613	83,804	83,804	91,463	98,569	105,653	111,656	111,656	115,571	119,176	122,260	124,729	124,729
TTM Pretax Income	8,272	14,859	25,031	37,745	37,745	52,970	65,391	75,874	85,313	85,313	93,299	100,819	108,535	115,268	115,268	119,928	124,327	128,235	131,550	131,550
TTM Net Income	7,635	13,083	21,647	32,312	32,312	44,837	55,049	63,648	70,897	70,897	77,527	83,680	90,084	95,672	95,672	99,541	103,192	106,435	109,186	109,186
TTM Dil. EPS	\$0.19	\$0.41	\$0.76	\$1.19	\$1.19	\$1.71	\$2.13	\$2.45	\$2.73	\$2.72	\$2.96	\$3.16	\$3.41	\$3.63	\$3.63	\$3.78	\$3.92	\$4.04	\$4.14	\$4.14
TTM Gross Profit	15,204	21,744	31,988	44,958	44,958	60,716	73,831	85,108	95,319	95,319	103,903	111,811	119,701	126,401	126,401	130,886	134,965	138,519	141,389	141,389
(Int Xp-Int Inc)/Shr	0.00	0.00	0.01	0.01	0.02	0.01	0.02	0.01	0.02	0.06	0.03	0.03	0.04	0.05	0.15	0.06	0.06	0.07	0.08	0.27
Net of Tax	0.00	0.00	0.01	0.01	0.02	0.01	0.01	0.01	0.02	0.05	0.02	0.03	0.03	0.04	0.12	0.05	0.05	0.06	0.07	0.23
Revenue % Change - Sequential	18.9%	87.8%	34.2%	22.0%	NM	17.8%	15.3%	10.3%	12.4%	NM	7.8%	5.2%	5.7%	3.2%	NM	(0.1%)	3.7%	5.2%	1.7%	NM
EPS % Change - Sequential	24.1%	147.5%	48.0%	28.3%	NM	18.7%	11.4%	8.8%	9.9%	NM	7.9%	5.7%	7.5%	4.1%	NM	(0.5%)	4.0%	5.0%	1.9%	NM
Revenue % Changes, yr. over yr.	(13.2%)	101.5%	205.5%	265.3%	125.9%	262.1%	122.4%	82.9%	68.5%	107.6%	54.1%	40.6%	34.7%	23.7%	36.9%	14.7%	13.0%	12.5%	10.9%	12.7%
EPS % Change, yr. over yr.	(19.7%)	425.2%	589.6%	487.5%	288.4%	461.9%	153.0%	84.7%	58.2%	120.1%	43.8%	36.3%	34.8%	27.6%	35.0%	17.7%	15.8%	13.1%	10.7%	14.2%
Gross Margin	66.8%	71.2%	75.0%	76.7%	73.8%	78.9%	75.7%	75.0%	73.0%	75.4%	72.6%	72.5%	73.4%	73.5%	73.0%	73.1%	72.8%	72.3%	71.9%	72.5%
R&D/Sales	18.6	10.7	8.7	7.9	10.0	7.6	7.5	7.2	6.9	7.3	6.8	6.8	6.8	6.8	6.8	7.0	6.9	6.8	6.8	6.9
SG&A/Sales	5.7	2.9	2.5	2.1	2.8	2.0	1.8	1.8	1.7	1.8	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Operating Expenses	24.3	13.6	11.2	10.0	12.8	9.6	9.3	9.0	8.7	9.1	8.5	8.5	8.5	8.5	8.5	8.7	8.6	8.5	8.5	8.5
Operating Margin	42.4	57.6	63.8	66.7	61.0	69.3	66.4	66.0	64.3	66.3	64.1	64.0	64.9	65.0	64.5	64.4	64.2	63.8	63.5	63.9
Non-Operating Expenses/Sales	3.0	1.9	1.6	1.6	1.8	1.6	1.7	1.4	1.6	1.6	1.9	2.2	2.5	2.9	2.4	3.2	3.6	3.8	4.3	3.7
Pretax Margin	43.6	58.5	64.7	67.8	62.0	70.5	67.6	67.0	65.6	67.5	65.6	65.9	67.1	67.6	66.6	67.4	67.5	67.4	67.5	67.4
Tax Rate	13.5	14.6	14.6	14.3	14.4	17.0	16.6	17.0	17.0	16.9	17.0	17.0	17.0	17.0	17.0	17.0	17.0	17.0	17.0	17.0
Net Margin	37.7	49.9	55.3	58.1	53.0	58.5	56.4	55.6	54.4	56.1	54.5	54.7	55.7	56.1	55.3	55.9	56.0	55.9	56.0	56.0

Source : Company data, Deutsche Bank estimates



28 August 2024
Semiconductors
NVIDIA Corporation



Appendix 1

Important Disclosures

*Other information available upon request

Disclosure checklist			
Company	Ticker	Recent price*	Disclosure
NVIDIA Corporation	NVDA.OQ	128.3 (USD) 27 Aug 2024	8, 15

*Prices are current as of the end of the previous trading session unless otherwise indicated and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Other information is sourced from Deutsche Bank, subject companies, and other sources. For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at <https://research.db.com/Research/Disclosures/EquityResearchDisclosures>. Aside from within this report, important risk and conflict disclosures can also be found at <https://research.db.com/Research/Disclosures/Disclaimer>. Investors are strongly encouraged to review this information before investing.

Important Disclosures Required by U.S. Regulators

Disclosures marked with an asterisk may also be required by at least one jurisdiction in addition to the United States. See Important Disclosures Required by Non-US Regulators and Explanatory Notes.

8. Deutsche Bank and/or its affiliate(s) expects to receive, or intends to seek, compensation for investment banking services from this company in the next three months.
15. This company has been a client of Deutsche Bank Securities Inc. within the past year during which time it received investment banking services.

Important Disclosures Required by Non-U.S. Regulators

Disclosures marked with an asterisk may also be required by at least one jurisdiction in addition to the United States. See Important Disclosures Required by Non-US Regulators and Explanatory Notes.

For disclosures pertaining to recommendations or estimates made on securities other than the primary subject of this research, please see the most recently published company report or visit our global disclosure look-up page on our website at <https://research.db.com/Research/Disclosures/EquityResearchDisclosures>. Aside from within this report, important risk and conflict disclosures can also be found at <https://research.db.com/Research/Disclosures/Disclaimer>. Investors are strongly encouraged to review this information before investing.

Analyst Certification

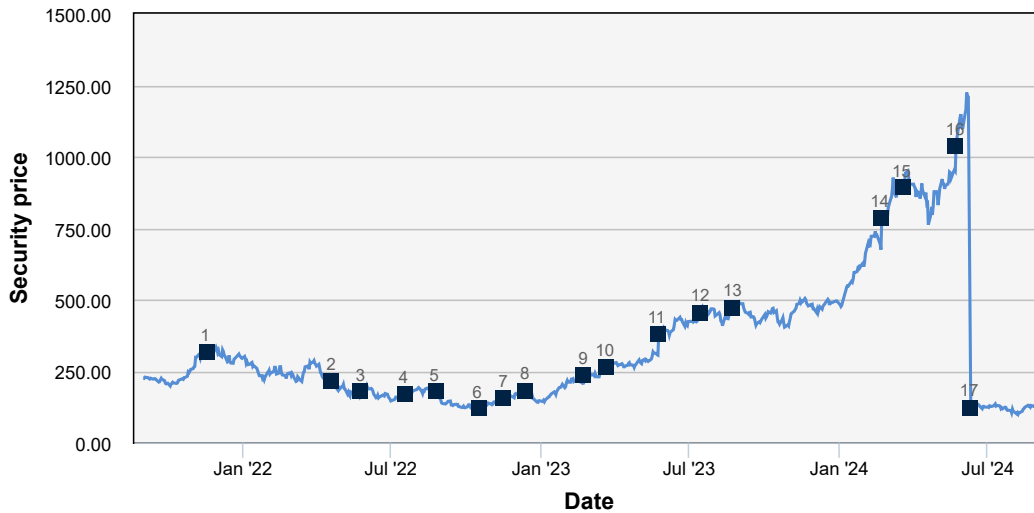
The views expressed in this report accurately reflect the personal views of the undersigned lead analyst(s) about the subject issuer and the securities of the issuer. In addition, the undersigned lead analyst(s) has not and will not receive any compensation for providing a specific recommendation or view in this report. Ross Seymore.

28 August 2024
Semiconductors
NVIDIA Corporation



Historical recommendations and target price: NVIDIA Corporation (NVDA.OQ)

(as of 08/28/2024)



Current Recommendations

Buy
Hold
Sell
Not Rated
Suspended Rating

** Analyst is no longer at Deutsche Bank

1.	11/18/2021	Hold, Target Price Change USD 285.00, Current Price USD 316.75 Ross Seymore	10.	03/22/2023	Hold, Target Price Change USD 220.00, Current Price USD 264.68 Ross Seymore
2.	04/20/2022	Hold, Target Price Change USD 255.00, Current Price USD 214.82 Ross Seymore	11.	05/25/2023	Hold, Target Price Change USD 390.00, Current Price USD 379.80 Ross Seymore
3.	05/26/2022	Hold, Target Price Change USD 190.00, Current Price USD 178.51 Ross Seymore	12.	07/16/2023	Hold, Target Price Change USD 440.00, Current Price USD 454.69 Ross Seymore
4.	07/19/2022	Hold, Target Price Change USD 175.00, Current Price USD 169.92 Ross Seymore	13.	08/24/2023	Hold, Target Price Change USD 560.00, Current Price USD 471.63 Ross Seymore
5.	08/25/2022	Hold, Target Price Change USD 165.00, Current Price USD 179.13 Ross Seymore	14.	02/22/2024	Hold, Target Price Change USD 720.00, Current Price USD 785.38 Ross Seymore
6.	10/17/2022	Hold, Target Price Change USD 140.00, Current Price USD 118.88 Ross Seymore	15.	03/19/2024	Hold, Target Price Change USD 850.00, Current Price USD 893.98 Ross Seymore
7.	11/17/2022	Hold, Target Price Change USD 150.00, Current Price USD 156.77 Ross Seymore	16.	05/23/2024	Hold, Target Price Change USD 1000.00, Current Price USD 1037.99 Ross Seymore
8.	12/13/2022	Hold, Target Price Change USD 170.00, Current Price USD 180.72 Ross Seymore	17.	06/10/2024	Hold, Target Price Change USD 100.00, Current Price USD 121.79 Joe Schwab
9.	02/23/2023	Hold, Target Price Change USD 200.00, Current Price USD 236.64 Ross Seymore			

Equity Rating Key

Buy: Based on a current 12-month view of TSR, we recommend that investors buy the stock.

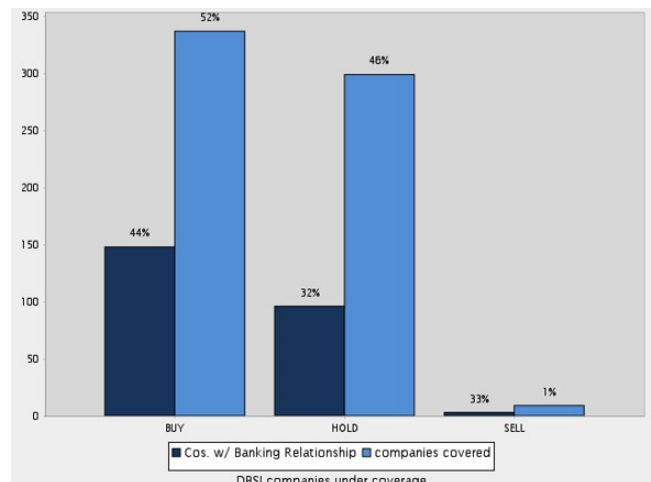
Sell: Based on a current 12-month view of TSR, we recommend that investors sell the stock.

Hold: We take a neutral view on the stock 12-months out and, based on this time horizon, do not recommend either a Buy or Sell.

TSR = Total Shareholder Return. Percentage change in share price from current price to projected target price plus projected dividend yield

Newly issued research recommendations and target prices supersede previously published research.

Equity rating dispersion and banking relationships



28 August 2024
Semiconductors
NVIDIA Corporation



Additional Information

The information and opinions in this report were prepared by Deutsche Bank AG or one of its affiliates (collectively 'Deutsche Bank'). Though the information herein is believed to be reliable and has been obtained from public sources believed to be reliable, Deutsche Bank makes no representation as to its accuracy or completeness. Hyperlinks to third-party websites in this report are provided for reader convenience only. Deutsche Bank neither endorses the content nor is responsible for the accuracy or security controls of those websites.

Effective 13 October 2023, Deutsche Bank AG acquired Numis Corporation Plc and its subsidiaries (the "Numis Group"). Numis Securities Limited ("NSL") is a member of the Numis Group and a firm authorised and regulated by the Financial Conduct Authority (Firm Reference Number: 144822). Deutsche Bank AG provides clients with, amongst other services, Investment Research services. NSL provides clients with, amongst other services, non-independent research services.

During an initial integration process, the research departments of Deutsche Bank AG and NSL will remain operationally distinct. Consequently, disclosures relating to conflicts of interest that may exist for Deutsche Bank AG and/or its affiliates do not currently take into account the business and activities of the Numis Group. The conflicts of interest that may exist for the Numis Group, in relation to the provision of research, can be found on the Numis website at <https://www.numis.com/legal-and-regulatory/conditions-and-disclaimers-that-govern-research-contained-in-the-research-pages-of-this-website>. The disclosures on this Numis webpage do not currently take into account the business and activities of Deutsche Bank AG and/or its affiliates which are not members of the Numis Group.

Additionally, any detailed conflicts of interest disclosures pertaining to a specific recommendation or estimate made on a security mentioned in this report or which have been included in our most recently published company report or found on our global disclosure look-up page, do not currently take into account the business and activities of the Numis Group. Instead, details of detailed conflicts of interest disclosures for the Numis Group, relating to specific issuers or securities, can be found at: https://library.numis.com/regulatory_notice. The issuer/security-specific conflict of interest disclosures on this Numis webpage do not take into account the business and activities of Deutsche Bank and/or its affiliates which are not members of the Numis Group.

If you use the services of Deutsche Bank in connection with a purchase or sale of a security that is discussed in this report, or is included or discussed in another communication (oral or written) from a Deutsche Bank analyst, Deutsche Bank may act as principal for its own account or as agent for another person.

Deutsche Bank may consider this report in deciding to trade as principal. It may also engage in transactions, for its own account or with customers, in a manner inconsistent with the views taken in this research report. Others within Deutsche Bank, including strategists, sales staff and other analysts, may take views that are inconsistent with those taken in this research report. Deutsche Bank issues a variety of research products, including fundamental analysis, equity-linked analysis, quantitative analysis and trade ideas. Recommendations contained in one type of communication may differ from recommendations contained in others, whether as a result of differing time horizons, methodologies, perspectives or otherwise. Deutsche Bank and/or its affiliates may also be holding debt or equity securities of the issuers it writes on. Analysts are paid in part based on the profitability of Deutsche Bank AG and its affiliates, which includes investment banking, trading and principal trading revenues.

Opinions, estimates and projections constitute the current judgment of the author as of the date of this report. They do not necessarily reflect the opinions of Deutsche Bank and are subject to change without notice. Deutsche Bank provides liquidity for buyers and sellers of securities issued by the companies it covers. Deutsche Bank research analysts sometimes have shorter-term trade ideas that may be inconsistent with Deutsche Bank's existing longer-term ratings. Some trade ideas for equities are listed as Catalyst Calls on the Research Website (<https://research.db.com/Research/>), and can be found on the general coverage list and also on the covered company's page. A Catalyst Call represents a high-conviction belief by an analyst that a stock will outperform or underperform the market and/or a specified sector over a time frame of no less than two weeks and no more than three months. In addition to Catalyst Calls, analysts may occasionally discuss with our clients, and with Deutsche Bank salespersons and traders, trading strategies or ideas that reference catalysts or events that may have a near-term or medium-term impact on the market price of the securities discussed in this report, which impact may be directionally counter to the analysts' current 12-month view of total return or investment return as described herein. Deutsche Bank has no obligation to update, modify or amend this report or to otherwise notify a recipient thereof if an opinion, forecast or estimate changes or becomes inaccurate. Coverage and the frequency of changes in market conditions and in both general and company-specific economic prospects make it difficult to update research at defined intervals. Updates are at the sole discretion of the coverage analyst or of the Research Department Management, and the majority of reports are published at irregular intervals. This report is provided for informational purposes only and does not take into account the particular investment objectives, financial situations, or needs of individual clients. It is not an offer or a solicitation of an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Target prices are inherently imprecise and a product of the analyst's judgment. The financial instruments discussed in this report may not be suitable for all investors, and investors must make their own informed investment decisions. Prices and availability of financial instruments are subject to change without notice, and investment transactions can lead to losses as a result of price fluctuations and other factors. If a financial instrument is denominated in a currency other than an investor's currency, a change in exchange rates may adversely affect the investment. Past performance is not necessarily indicative of future results. Performance calculations exclude transaction costs, unless otherwise indicated. Unless otherwise indicated, prices are current as of the end of the previous trading session and are sourced from local exchanges via Reuters, Bloomberg and other vendors. Data is also sourced from Deutsche Bank, subject companies, and other parties.

28 August 2024
Semiconductors
NVIDIA Corporation



The Deutsche Bank Research Department is independent of other business divisions of the Bank. Details regarding our organizational arrangements and information barriers we have to prevent and avoid conflicts of interest with respect to our research are available on our website (<https://research.db.com/Research/>) under Disclaimer.

Macroeconomic fluctuations often account for most of the risks associated with exposures to instruments that promise to pay fixed or variable interest rates. For an investor who is long fixed-rate instruments (thus receiving these cash flows), increases in interest rates naturally lift the discount factors applied to the expected cash flows and thus cause a loss. The longer the maturity of a certain cash flow and the higher the move in the discount factor, the higher will be the loss. Upside surprises in inflation, fiscal funding needs, and FX depreciation rates are among the most common adverse macroeconomic shocks to receivers. But counterparty exposure, issuer creditworthiness, client segmentation, regulation (including changes in assets holding limits for different types of investors), changes in tax policies, currency convertibility (which may constrain currency conversion, repatriation of profits and/or liquidation of positions), and settlement issues related to local clearing houses are also important risk factors. The sensitivity of fixed-income instruments to macroeconomic shocks may be mitigated by indexing the contracted cash flows to inflation, to FX depreciation, or to specified interest rates - these are common in emerging markets. The index fixings may - by construction - lag or mis-measure the actual move in the underlying variables they are intended to track. The choice of the proper fixing (or metric) is particularly important in swaps markets, where floating coupon rates (i.e., coupons indexed to a typically short-dated interest rate reference index) are exchanged for fixed coupons. Funding in a currency that differs from the currency in which coupons are denominated carries FX risk. Options on swaps (swaptions) the risks typical to options in addition to the risks related to rates movements.

Derivative transactions involve numerous risks including market, counterparty default and illiquidity risk. The appropriateness of these products for use by investors depends on the investors' own circumstances, including their tax position, their regulatory environment and the nature of their other assets and liabilities; as such, investors should take expert legal and financial advice before entering into any transaction similar to or inspired by the contents of this publication. The risk of loss in futures trading and options, foreign or domestic, can be substantial. As a result of the high degree of leverage obtainable in futures and options trading, losses may be incurred that are greater than the amount of funds initially deposited - up to theoretically unlimited losses. Trading in options involves risk and is not suitable for all investors. Prior to buying or selling an option, investors must review the 'Characteristics and Risks of Standardized Options', at <https://www.theocc.com/company-information/documents-and-archives/publications>. If you are unable to access the website, please contact your Deutsche Bank representative for a copy of this important document.

Participants in foreign exchange transactions may incur risks arising from several factors, including the following: (i) exchange rates can be volatile and are subject to large fluctuations; (ii) the value of currencies may be affected by numerous market factors, including world and national economic, political and regulatory events, events in equity and debt markets and changes in interest rates; and (iii) currencies may be subject to devaluation or government-imposed exchange controls, which could affect the value of the currency. Investors in securities such as ADRs, whose values are affected by the currency of an underlying security, effectively assume currency risk.

Unless governing law provides otherwise, all transactions should be executed through the Deutsche Bank entity in the investor's home jurisdiction. Aside from within this report, important conflict disclosures can also be found at <https://research.db.com/Research/> on each company's research page. Investors are strongly encouraged to review this information before investing.

Deutsche Bank (which includes Deutsche Bank AG, its branches and affiliated companies) is not acting as a financial adviser, consultant or fiduciary to you or any of your agents (collectively, "You" or "Your") with respect to any information provided in this report. Deutsche Bank does not provide investment, legal, tax or accounting advice, Deutsche Bank is not acting as your impartial adviser, and does not express any opinion or recommendation whatsoever as to any strategies, products or any other information presented in the materials. Information contained herein is being provided solely on the basis that the recipient will make an independent assessment of the merits of any investment decision, and it does not constitute a recommendation of, or express an opinion on, any product or service or any trading strategy.

The information presented is general in nature and is not directed to retirement accounts or any specific person or account type, and is therefore provided to You on the express basis that it is not advice, and You may not rely upon it in making Your decision. The information we provide is being directed only to persons we believe to be financially sophisticated, who are capable of evaluating investment risks independently, both in general and with regard to particular transactions and investment strategies, and who understand that Deutsche Bank has financial interests in the offering of its products and services. If this is not the case, or if You are an IRA or other retail investor receiving this directly from us, we ask that you inform us immediately.

In July 2018, Deutsche Bank revised its rating system for short term ideas whereby the branding has been changed to Catalyst Calls ("CC") from SOLAR ideas; the rating categories for Catalyst Calls originated in the Americas region have been made consistent with the categories used by Analysts globally; and the effective time period for CCs has been reduced from a maximum of 180 days to 90 days.

United States: Approved and/or distributed by Deutsche Bank Securities Incorporated, a member of FINRA and SIPC. Analysts located outside of the United States are employed by non-US affiliates and are not registered/qualified as research analysts with FINRA.

European Economic Area (exc. United Kingdom): Approved and/or distributed by Deutsche Bank AG, a joint stock corporation with limited liability incorporated in the Federal Republic of Germany with its principal office in Frankfurt am Main.

28 August 2024
Semiconductors
NVIDIA Corporation



Deutsche Bank AG is authorized under German Banking Law and is subject to supervision by the European Central Bank and by BaFin, Germany's Federal Financial Supervisory Authority. This research report is also distributed by Numis Europe Limited ('NEL'), to EEA clients as third-party research. NEL is an investment firm authorised and regulated by the Central Bank of Ireland ('CBI'), with its registered address at Riverview House, 21-23 City Quay, Dublin 2, D02 F21, The Republic of Ireland.

United Kingdom: Approved and/or distributed by Deutsche Bank AG acting through its London Branch at 21 Moorfields, London EC2Y 9DB. Deutsche Bank AG in the United Kingdom is authorised by the Prudential Regulation Authority and is subject to limited regulation by the Prudential Regulation Authority and Financial Conduct Authority. Details about the extent of our authorisation and regulation are available on request.

Hong Kong SAR: Distributed by Deutsche Bank AG, Hong Kong Branch except for any research content relating to futures contracts within the meaning of the Hong Kong Securities and Futures Ordinance Cap. 571. Research reports on such futures contracts are not intended for access by persons who are located, incorporated, constituted or resident in Hong Kong. The author(s) of a research report may not be licensed to carry on regulated activities in Hong Kong and, if not licensed, do not hold themselves out as being able to do so. The provisions set out above in the 'Additional Information' section shall apply to the fullest extent permissible by local laws and regulations, including without limitation the Code of Conduct for Persons Licensed or Registered with the Securities and Futures Commission. This report is intended for distribution only to 'professional investors' as defined in Part 1 of Schedule of the SFO. This document must not be acted or relied on by persons who are not professional investors. Any investment or investment activity to which this document relates is only available to professional investors and will be engaged only with professional investors.

India: Prepared by Deutsche Equities India Private Limited (DEIPL) having CIN: U65990MH2002PTC137431 and registered office at 14th Floor, The Capital, C-70, G Block, Bandra Kurla Complex, Mumbai (India) 400051. Tel: + 91 22 7180 4444. It is registered by the Securities and Exchange Board of India (SEBI) as a Stock broker bearing registration no.: INZ000252437; Merchant Banker bearing SEBI Registration no.: INM000010833 and Research Analyst bearing SEBI Registration no.: INH000001741. DEIPL's Compliance / Grievance officer is Ms. Rashmi Poddar (Tel: +91 22 7180 4929 email ID: complaints.deipl@db.com). Registration granted by SEBI and certification from NISM in no way guarantee performance of DEIPL or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. DEIPL may have received administrative warnings from the SEBI for breaches of Indian regulations. Deutsche Bank and/or its affiliate(s) may have debt holdings or positions in the subject company. With regard to information on associates, please refer to the "Shareholdings" section in the Annual Report at: <https://www.db.com/ir/en/annual-reports.htm>.

Japan: Approved and/or distributed by Deutsche Securities Inc.(DSI). Registration number - Registered as a financial instruments dealer by the Head of the Kanto Local Finance Bureau (Kinsho) No. 117. Member of associations: JSDA, Type II Financial Instruments Firms Association and The Financial Futures Association of Japan. Commissions and risks involved in stock transactions - for stock transactions, we charge stock commissions and consumption tax by multiplying the transaction amount by the commission rate agreed with each customer. Stock transactions can lead to losses as a result of share price fluctuations and other factors. Transactions in foreign stocks can lead to additional losses stemming from foreign exchange fluctuations. We may also charge commissions and fees for certain categories of investment advice, products and services. Recommended investment strategies, products and services carry the risk of losses to principal and other losses as a result of changes in market and/or economic trends, and/or fluctuations in market value. Before deciding on the purchase of financial products and/or services, customers should carefully read the relevant disclosures, prospectuses and other documentation. 'Moody's', 'Standard Poor's', and 'Fitch' mentioned in this report are not registered credit rating agencies in Japan unless Japan or 'Nippon' is specifically designated in the name of the entity. Reports on Japanese listed companies not written by analysts of DSI are written by Deutsche Bank Group's analysts with the coverage companies specified by DSI. Some of the foreign securities stated on this report are not disclosed according to the Financial Instruments and Exchange Law of Japan. Target prices set by Deutsche Bank's equity analysts are based on a 12-month forecast period.

Korea: Distributed by Deutsche Securities Korea Co.

South Africa: Deutsche Bank AG Johannesburg is incorporated in the Federal Republic of Germany (Branch Register Number in South Africa: 1998/003298/10).

Singapore: This report is issued by Deutsche Bank AG, Singapore Branch (One Raffles Quay #18-00 South Tower Singapore 048583, 65 6423 8001), which may be contacted in respect of any matters arising from, or in connection with, this report. Where this report is issued or promulgated by Deutsche Bank in Singapore to a person who is not an accredited investor, expert investor or institutional investor (as defined in the applicable Singapore laws and regulations), they accept legal responsibility to such person for its contents.

Taiwan: Information on securities/investments that trade in Taiwan is for your reference only. Readers should independently evaluate investment risks and are solely responsible for their investment decisions. Deutsche Bank research may not be distributed to the Taiwan public media or quoted or used by the Taiwan public media without written consent. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation to trade in such securities/instruments.

Qatar: Deutsche Bank AG in the Qatar Financial Centre (registered no. 00032) is regulated by the Qatar Financial Centre Regulatory Authority. Deutsche Bank AG - QFC Branch may undertake only the financial services activities that fall within the scope of its existing QFCRA license. Its principal place of business in the QFC: Qatar Financial Centre, Tower, West Bay, Level

28 August 2024
Semiconductors
NVIDIA Corporation



5, PO Box 14928, Doha, Qatar. This information has been distributed by Deutsche Bank AG. Related financial products or services are only available only to Business Customers, as defined by the Qatar Financial Centre Regulatory Authority.

Russia: The information, interpretation and opinions submitted herein are not in the context of, and do not constitute, any appraisal or evaluation activity requiring a license in the Russian Federation.

Kingdom of Saudi Arabia: Deutsche Securities Saudi Arabia (DSSA) is a closed joint stock company authorized by the Capital Market Authority of the Kingdom of Saudi Arabia with a license number (No. 37-07073) to conduct the following business activities: Dealing, Arranging, Advising, and Custody activities. DSSA registered office is Faisaliah Tower, 17th Floor, King Fahad Road - Al Olaya District Riyadh, Kingdom of Saudi Arabia P.O. Box 301806.

United Arab Emirates: Deutsche Bank AG in the Dubai International Financial Centre (registered no. 00045) is regulated by the Dubai Financial Services Authority. Deutsche Bank AG - DIFC Branch may only undertake the financial services activities that fall within the scope of its existing DFSA license. Principal place of business in the DIFC: Dubai International Financial Centre, The Gate Village, Building 5, PO Box 504902, Dubai, U.A.E. This information has been distributed by Deutsche Bank AG. Related financial products or services are available only to Professional Clients, as defined by the Dubai Financial Services Authority.

Australia and New Zealand: This research is intended only for 'wholesale clients' within the meaning of the Australian Corporations Act and New Zealand Financial Advisors Act, respectively. Please refer to Australia-specific research disclosures and related information at https://www.dbresearch.com/PROD/RPS_EN-PROD/PROD0000000000521304.xhtml. Where research refers to any particular financial product recipients of the research should consider any product disclosure statement, prospectus or other applicable disclosure document before making any decision about whether to acquire the product. In preparing this report, the primary analyst or an individual who assisted in the preparation of this report has likely been in contact with the company that is the subject of this research for confirmation/clarification of data, facts, statements, permission to use company-sourced material in the report, and/or site-visit attendance. Without prior approval from Research Management, analysts may not accept from current or potential Banking clients the costs of travel, accommodations, or other expenses incurred by analysts attending site visits, conferences, social events, and the like. Similarly, without prior approval from Research Management and Anti-Bribery and Corruption ("ABC") team, analysts may not accept perks or other items of value for their personal use from issuers they cover.

Additional information relative to securities, other financial products or issuers discussed in this report is available upon request. This report may not be reproduced, distributed or published without Deutsche Bank's prior written consent.

Backtested, hypothetical or simulated performance results have inherent limitations. Unlike an actual performance record based on trading actual client portfolios, simulated results are achieved by means of the retroactive application of a backtested model itself designed with the benefit of hindsight. Taking into account historical events the backtesting of performance also differs from actual account performance because an actual investment strategy may be adjusted any time, for any reason, including a response to material, economic or market factors. The backtested performance includes hypothetical results that do not reflect the reinvestment of dividends and other earnings or the deduction of advisory fees, brokerage or other commissions, and any other expenses that a client would have paid or actually paid. No representation is made that any trading strategy or account will or is likely to achieve profits or losses similar to those shown. Alternative modeling techniques or assumptions might produce significantly different results and prove to be more appropriate. Past hypothetical backtest results are neither an indicator nor guarantee of future returns. Actual results will vary, perhaps materially, from the analysis.

The method for computing individual E,S,G and composite ESG scores set forth herein is a novel method developed by the Research department within Deutsche Bank AG, computed using a systematic approach without human intervention. Different data providers, market sectors and geographies approach ESG analysis and incorporate the findings in a variety of ways. As such, the ESG scores referred to herein may differ from equivalent ratings developed and implemented by other ESG data providers in the market and may also differ from equivalent ratings developed and implemented by other divisions within the Deutsche Bank Group. Such ESG scores also differ from other ratings and rankings that have historically been applied in research reports published by Deutsche Bank AG. Further, such ESG scores do not represent a formal or official view of Deutsche Bank AG. It should be noted that the decision to incorporate ESG factors into any investment strategy may inhibit the ability to participate in certain investment opportunities that otherwise would be consistent with your investment objective and other principal investment strategies. The returns on a portfolio consisting primarily of sustainable investments may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered, and the investment opportunities available to such portfolios may differ. Companies may not necessarily meet high performance standards on all aspects of ESG or sustainable investing issues; there is also no guarantee that any company will meet expectations in connection with corporate responsibility, sustainability, and/or impact performance.

Copyright © 2024 Deutsche Bank AG



David Folkerts-Landau

Group Chief Economist and Global Head of Research

Pam Finelli Global Chief Operating Officer Research	Steve Pollard Global Head of Company Research and Sales	Jim Reid Global Head of Macro and Thematic Research	Tim Rokossa Head of Germany Research
Gerry Gallagher Head of European Company Research	Matthew Barnard Head of Americas Company Research	Peter Milliken Head of APAC Company Research	Debbie Jones Global Head of Sustainability, Research
Sameer Goel Global Head of EM & APAC Research	Francis Yared Global Head of Rates Research	George Saravelos Global Head of FX Research	Peter Hooper Vice-Chair of Research

International Production Locations

Deutsche Bank AG Deutsche Bank Place Level 16 Corner of Hunter & Phillip Streets Sydney, NSW 2000 Australia Tel: (61) 2 8258 1234	Deutsche Bank AG Equity Research Mainzer Landstrasse 11-17 60329 Frankfurt am Main Germany Tel: (49) 69 910 00	Deutsche Bank AG Filiale Hongkong International Commerce Centre, 1 Austin Road West, Kowloon, Hong Kong Tel: (852) 2203 8888	Deutsche Securities Inc. 1-3-1 Azabudai Azabudai Hills Mori JP Tower Minato-ku, Tokyo 106-0041 Japan Tel: (81) 3 6730 1000
Deutsche Bank AG 21 Moorfields London EC2Y 9DB United Kingdom Tel: (44) 20 7545 8000	Deutsche Bank Securities Inc. The Deutsche Bank Center 1 Columbus Circle New York, NY 10019 Tel: (1) 212 250 2500		